

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Explainable Retinal-Clinical Adaptive Fusion Network (RCAFN) for Early Cardiovascular Disease Risk Stratification

Abstract

Physicians assessing cardiovascular disease (CVD) risk in asymptomatic patients often lack an accessible, low-cost first-line screening tool, especially in primary-care and ophthalmic settings. Standard risk-stratification tests — CT coronary calcium scoring, echocardiography, and lipid panels — require dedicated equipment, radiation, or invasive sampling, and cost significantly more than a routine eye exam, making population-scale screening impractical, particularly in resource-limited settings.

At present, clinicians rely on symptom presentation or referral-based cardiology workups, a reactive process that often catches disease only after significant atherosclerotic progression. Large cohort studies have linked retinal microvascular changes — arteriolar narrowing, venular widening, altered artery-to-vein ratio — to future cardiovascular events, and a 2024 scoping review of 24 studies confirmed deep learning can extract this risk signal from fundus photographs, with one such model already approved as a medical device internationally. However, the same review found most existing models are black boxes, offering a risk score without clinical interpretability — limiting real-world trust and adoption.

Lee et al. (2023) showed image-clinical fusion improves CVD prediction, and Guo et al. (2025) provided a public dataset for it — but neither validates the model's explainability against clinical ground truth, the gap this project addresses. Building on these gaps, this project proposes the Retinal-Clinical Adaptive Fusion Network (RCAFN) — a ResNet-50-based architecture that encodes CLAHE-preprocessed fundus images, fuses them with clinical variables (age, gender), and outputs a cardiovascular risk classification alongside a Grad-CAM explanation cross-checked against clinically documented CVD-relevant retinal features.

The project is scoped to cardiovascular risk classification using carotid intima-media thickness (CIMT) as a clinical proxy label, using the China-Fundus-CIMT dataset (2,903 patients), supplemented by EyePACS/APTOS for pretraining. It excludes a full clinical risk-factor panel (BP, cholesterol, smoking — absent from the dataset), differential diagnosis, real-time deployment, and hospital-system integration. Success is measured by the RCAFN's multimodal configuration exceeding the best image-only baseline's AUC-ROC by a measurable margin, and by Grad-CAM explanations being judged consistent with established retinal biomarkers in an independent qualitative review.

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References (Category specific)

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4. "Noninvasive Coronary Artery Disease Detection Using Retinal Images: A Multimodal Study." JACC: Advances, 2025. <https://doi.org/10.1016/j.jacadv.2025.102341>.

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